

RippleWealth Deployment Manual

RippleWealth is an AI-powered portfolio analytics platform integrating real-time stock data, AI-driven risk analysis, portfolio optimization, and cloud deployment architecture.

Project Architecture

Frontend: Static HTML/CSS/JavaScript deployed on Netlify

Backend: FastAPI API deployed on Railway

Database: SQLite

Market Data Provider: yFinance

Project Structure

RippleWealth/

```
|
├── frontend/
│   ├── index.html
│   ├── dashboard.html
│   ├── login.html
│   ├── register.html
│   ├── assets/
│   ├── styles/
│   └── scripts/
├── backend/
│   ├── main.py
│   ├── requirements.txt
│   ├── Procfile
│   ├── ripple.db
│   └── database models
├── ai_layer/
│   ├── agent_orchestrator.py
│   ├── data_agent.py
│   ├── risk_agent.py
│   ├── optimization_agent.py
│   ├── simulation_agent.py
│   ├── recommendation_agent.py
│   ├── scenario_agent.py
│   ├── explanation_agent.py
│   └── demo.py
└── deployment/
    ├── Railway (Backend)
```

└─ Netlify (Frontend)

Backend Deployment using Railway

- Install Railway CLI using: `npm install -g @railway/cli`
- Login to Railway using: `railway login`
- Navigate to backend folder using: `cd backend`
- Create requirements.txt containing FastAPI and required dependencies.
- Create Procfile containing: `web: uvicorn main:app --host 0.0.0.0 --port $PORT`
- Deploy backend using: `railway up`
- Generate a public Railway domain from Networking settings.
- Access FastAPI Swagger documentation using `/docs` endpoint.

Important Railway Notes

Opening the Railway root URL may display:

```
{"detail": "Not Found"}
```

This is normal because FastAPI does not contain a root endpoint. API documentation is available under the `/docs` route.

Example:

`https://your-railway-url.up.railway.app/docs`

Frontend Deployment using Netlify

- Navigate to frontend folder.
- Replace localhost backend URLs with Railway production URL.
- Create netlify.toml file with `publish='.'` configuration.
- Zip the entire frontend folder.
- Open Netlify Drop deployment page.
- Drag and drop the frontend folder into Netlify.
- Netlify generates a public deployment URL.

Frontend Deployment Notes

RippleWealth is a multi-page static HTML application. Pages are accessed directly using:

`/dashboard.html`

`/login.html`

`/register.html`

Implemented Features

- User Authentication
- Portfolio Management
- Live Market Price Integration
- Portfolio Allocation Engine
- Sharpe Ratio Analysis
- Value at Risk (VaR)
- AI-based Portfolio Recommendations
- Scenario Simulation
- Interactive Financial Dashboard

Production Recommendations

- Upgrade SQLite to PostgreSQL
- Add JWT Authentication
- Implement Redis Caching
- Use WebSockets for Live Streaming
- Containerize using Docker
- Implement CI/CD Pipelines

Conclusion

RippleWealth demonstrates a complete cloud-hosted AI-powered fintech platform with scalable backend architecture, real-time market integration, and interactive portfolio analytics.

App link - <https://ripplewealth.netlify.app/>